

Merchanta AI — Decision Log

Project

Merchanta AI

Purpose

This log outlines significant architectural, product, technical, and implementation choices made throughout the Merchanta AI development process.

1. Product Direction Decision

Field	Details
Decision ID	DL-001
Decision	Build an AI commerce optimization platform
Status	Approved
Reason	AI shopping assistants are becoming major discovery channels
Impact	Defines core platform positioning
Alternatives Considered	Traditional SEO analytics platform
Final Outcome	The focus shifted toward AI-native commerce optimization

2. Ecommerce Platform Decision

Field	Details
Decision ID	DL-002
Decision	Use Shopify as the initial e-commerce integration
Status	Approved
Reason	Shopify has strong APIs and large merchant adoption
Impact	Simplifies onboarding and integration

Alternatives Considered	WooCommerce, Magento
Final Outcome	Shopify-first architecture implemented

3. Frontend Framework Decision

Field	Details
Decision ID	DL-003
Decision	Use Next.js with TypeScript
Status	Approved
Reason	Better scalability, routing, performance, and developer experience
Impact	Enables modular frontend architecture
Alternatives Considered	React SPA, Vue.js
Final Outcome	Next.js + TypeScript selected

4. Styling Framework Decision

Field	Details
Decision ID	DL-004
Decision	Use Tailwind CSS
Status	Approved
Reason	Faster UI development and responsive styling
Impact	Consistent design system
Alternatives Considered	Bootstrap, Material UI
Final Outcome	Tailwind CSS adopted

5. Backend Architecture Decision

Field	Details
Decision ID	DL-005

Decision	Use Node.js + Express.js backend
Status	Approved
Reason	Efficient REST API handling and ecosystem maturity
Impact	Simplifies API orchestration
Alternatives Considered	Django, NestJS
Final Outcome	Express.js architecture selected

6. Database Selection Decision

Field	Details
Decision ID	DL-006
Decision	Use MongoDB Atlas
Status	Approved
Reason	Flexible schema structure for dynamic e-commerce data
Impact	Faster iteration and easier metadata storage
Alternatives Considered	PostgreSQL, MySQL
Final Outcome	MongoDB Atlas selected

7. AI Service Architecture Decision

Field	Details
Decision ID	DL-007
Decision	Separate AI engine into independent Python service
Status	Approved
Reason	AI workloads require independent scaling
Impact	Enables modular AI infrastructure
Alternatives Considered	Single monolithic backend
Final Outcome	FastAPI AI microservice implemented

8. AI Framework Decision

Field	Details
Decision ID	DL-008
Decision	Use FastAPI for AI services.
Status	Approved
Reason	High-performance async APIs for AI pipelines
Impact	Improves AI request handling
Alternatives Considered	Flask, Django
Final Outcome	FastAPI selected

9. Semantic Search Decision

Field	Details
Decision ID	DL-009
Decision	Use embeddings + vector retrieval
Status	Approved
Reason	AI discoverability requires semantic understanding
Impact	Enables contextual product matching
Alternatives Considered	Keyword-based search only
Final Outcome	Embedding-based retrieval implemented

10. Vector Database Decision

Field	Details
Decision ID	DL-010
Decision	Use Pinecone / Weaviate
Status	Approved
Reason	Efficient similarity search and semantic retrieval
Impact	Enables scalable RAG architecture

Alternatives Considered	Elasticsearch
Final Outcome	Vector database architecture adopted

11. AI Architecture Decision

Field	Details
Decision ID	DL-011
Decision	Use Retrieval-Augmented Generation (RAG)
Status	Approved
Reason	Improves contextual AI evaluation and recommendation quality
Impact	Enhances AI reasoning accuracy
Alternatives Considered	Direct prompt-only architecture
Final Outcome	RAG pipelines integrated

12. Simulation Feature Decision

Field	Details
Decision ID	DL-012
Decision	Build conversational shopping simulation engine
Status	Approved
Reason	Merchants need explainable AI ranking insights
Impact	Creates highly interactive merchant experience
Alternatives Considered	Static analytics dashboard
Final Outcome	Simulation engine prioritized

13. AI Visibility Scoring Decision

Field	Details
Decision ID	DL-013

Decision	Create unified AI visibility score
Status	Approved
Reason	Merchants need measurable AI discoverability metrics
Impact	Simplifies performance interpretation
Alternatives Considered	Separate fragmented metrics
Final Outcome	Unified scoring framework adopted

14. Recommendation System Decision

Field	Details
Decision ID	DL-014
Decision	Generate actionable optimization recommendations
Status	Approved
Reason	Merchants require implementation guidance, not only analytics
Impact	Improves practical business value
Alternatives Considered	Analytics-only reporting
Final Outcome	AI-generated recommendations included

15. Queue Processing Decision

Field	Details
Decision ID	DL-015
Decision	Use BullMQ + Redis for background jobs
Status	Approved
Reason	AI processing requires asynchronous execution
Impact	Improves scalability and responsiveness
Alternatives Considered	Synchronous processing
Final Outcome	Queue-based architecture adopted

16. Authentication Decision

Field	Details
Decision ID	DL-016
Decision	Use JWT authentication
Status	Approved
Reason	Lightweight and scalable authentication
Impact	Simplifies API security
Alternatives Considered	Session-based auth
Final Outcome	JWT authentication implemented

17. Hosting Architecture Decision

Field	Details
Decision ID	DL-017
Decision	Separate hosting by service layer
Status	Approved
Reason	Different workloads require independent scaling
Impact	Better infrastructure flexibility
Alternatives Considered	Single-server deployment
Final Outcome	Distributed cloud deployment selected

18. Frontend Hosting Decision

Field	Details
Decision ID	DL-018
Decision	Use Vercel for frontend deployment
Status	Approved
Reason	Native Next.js optimization
Impact	Faster deployment and CDN delivery

Alternatives Considered	Netlify
Final Outcome	Vercel selected

19. Monitoring Decision

Field	Details
Decision ID	DL-019
Decision	Integrate centralized monitoring tools
Status	Approved
Reason	AI systems require observability and debugging
Impact	Better reliability and issue tracking
Alternatives Considered	Minimal logging setup
Final Outcome	Sentry + Grafana + Prometheus selected

20. Product Positioning Decision

Field	Details
Decision ID	DL-020
Decision	Position platform as “SEO for AI Shopping Agents”
Status	Approved
Reason	Clearly communicates business value
Impact	Stronger product differentiation
Alternatives Considered	Generic AI analytics platform
Final Outcome	AI-commerce optimization positioning finalized

21. Future Expansion Decision

Field	Details
Decision ID	DL-021

Decision	Design architecture for multi-platform e-commerce support
Status	Planned
Reason	Future scalability across e-commerce ecosystems
Impact	Easier future integrations
Alternatives Considered	Shopify-only tightly coupled design
Final Outcome	Modular integration architecture planned

22. Summary

These decisions have shaped Merchanta AI into the following:

- An AI-native e-commerce intelligence platform
- A scalable, multi-service architecture
- A semantic AI recommendation optimization system
- A conversational commerce analytics platform

Together, these architectural and product choices provide a foundation for long-term scalability, AI readiness, and merchant-centric optimization.